Supplementary Materials

Our Supplementary Materials consist of two sections. In the first section, we provide a more detailed mathematical explanation of how our objective function J_{GIB} can be derived from first principles. In the second section, we explain key details relating to the operationalization of our model.

1. Deriving our Objective Function

At a high level, the standard Information Bottleneck (IB) framework can be broken down into two sub-problems. In the first sub-problem, the goal is to **compress** meanings M into words W. In the second, the goal is to **predict** the referents U from the words W. As explained in the main text, our novel objective function can also be framed as a combination of compression and prediction problems, albeit with a larger set of variables.

To account for this additional complexity in a theoretically-grounded manner, we leverage the Generalized IB (GIB) framework (Slonim, Friedman, & Tishby, 2006). GIB formalizes these problems of compression and prediction using Bayes networks, which can account for larger sets of variables given an appropriate set of probabilistic dependencies. Here, we first provide an intuition about the GIB framework by explaining how the standard IB formulation can be recovered using this approach. Then, we illustrate how our novel objective function can be derived from GIB.

1.1. *Deriving Standard IB from GIB*

In general, GIB requires two Bayes networks to be specified over the same set of variables. The first Bayes network, $G_{compress}$, models the compression sub-problem. The goal is to **minimize** the total information present in $G_{compress}$, as this leads to a greater degree of compression (we explain how to compute the total information in a Bayes network later in this section). The second Bayes network, $G_{predict}$, models the prediction sub-problem. The goal is to **maximize** the total information present in $G_{predict}$, as this ensures a higher degree of predictability between the variables.

As a concrete example, see Figure 1. Here, $G_{compress}$ specifies that both the referents U and words W are dependent on the meanings M. The arrow from M to U can be thought of as decoding a speaker's noisy mental representation

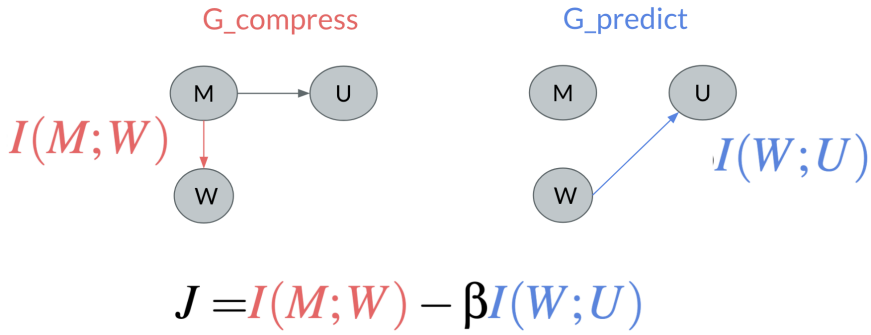

Figure 1. Visualization of Bayes networks for the standard IB problem. Terms in red correspond to information being minimized, and terms in blue correspond to information being maximized. The grey arrow corresponds to a constant relationship between variables M and U.

of a meaning m into a specific referent u. This decoding process relies on the distribution $p(u|m)$, which captures confusability between referents according to the structure of the semantic space. The semantic space is typically assumed to be universal in the standard IB formulation, and thus the relationship between M and U is constant and plays no role in optimizing the compression. The arrow from M to W can be thought of as compressing meanings into words using an encoder $q(w|m)$, which represents a language's lexical semantic system. Turning to $G_{predict}$, we see that a single arrow specifies that the referents U are predicted from words W. This models a pressure for expressivity, where words W should be maximally predictive of the true referent a speaker intended to communicate.

As mentioned earlier, GIB dictates that the total information of $G_{compress}$ should be minimized, and the total information of $G_{predict}$ should be maximized. The total information of a Bayes network is simply the sum of the mutual information terms between each variable and its parent (Slonim et al., 2006). For example, the red term to the left of $G_{compress}$ in Figure 1 shows the mutual information being minimized (recall that the mutual information between M and U is constant). Likewise, the blue term to the right of $G_{predict}$ in Figure 1 shows the mutual information being maximized. Combining the two terms into a single objective function gives rise to function J at the bottom of Figure 1, where the term β modulates the importance of predictability in the trade-off. This objective function is identical to J_{IB} in the main text, showing that we are able to recover the standard IB formulation using the GIB framework.

1.2. *Deriving Our Novel Objective Function from GIB*

We can now explain how our objective function can be derived from these same principles. As mentioned in the main text, we consider additional compression and prediction problems that capture higher-level clusterings of meanings according

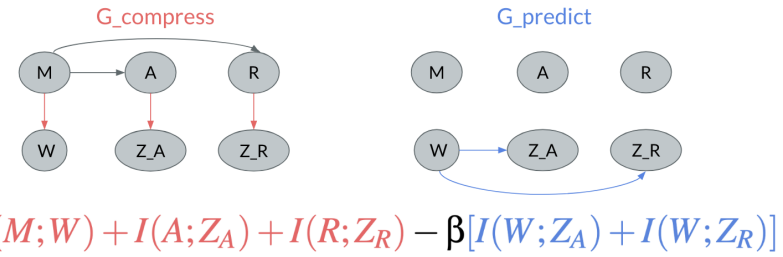

Figure 2. Visualization of Bayes networks for our extended formulation. Terms in red correspond to information being minimized, and terms in blue correspond to information being maximized. The grey arrows correspond to constant relationships.

to shared feature values on a particular dimension. In the domain of Shepard circles, we have two relevant feature dimensions, angle A and radius R, which are compressed into sets of higher-level features Z_A and Z_R respectively. For example, Language 3 in Figure 2 of the main body of the paper has a cluster SMALL that groups several values of radius into a single higher-level feature.

Figure 2 shows $G_{compress}$ and $G_{predict}$ for our new formulation. In $G_{compress}$, we see that the set of meanings M has a constant relationship with each of the feature dimensions. The compressions occur between meanings M and words W, as well as between each feature dimension, A and R, and their respective sets of high-level features, Z_A and Z_R. Thus, minimizing the total information of $G_{compress}$ requires minimizing the terms in red in the objective function J at the bottom of Figure 2. In $G_{predict}$, we see that the words W need to be predictive of the higher-level features Z_A and Z_R, as we motivated in the main text. Thus, maximizing the total information of $G_{predict}$ requires maximizing the terms in blue in Figure 2. This example shows how our objective function can be motivated from first principles.

Note that there is a slight difference between the objective function J in Figure 2 and J_{GIB} in the main text. Specifically, in Figure 2, J considers the mutual information terms $I(A; Z_A)$ and $I(R; Z_R)$, while in the main text, J_{GIB} considers the mutual information terms $I(M; Z_A)$ and $I(M; Z_R)$. The terms $I(A; Z_A)$ and $I(R; Z_R)$ are more precise; we replaced the individual feature dimension with meanings M in the main text to ease interpretation of our model. Since we assume deterministic mappings between meanings M and each of the feature dimensions, it becomes the case that $I(A; Z_A) = I(M; Z_A)$ and $I(R; Z_R) = I(M; Z_R)$, and thus, the two versions of the objective function are identical in practice.

2. Operationalization Details

Both our model and the standard IB formulation require specifying conditional distributions that define the relationships between the variables in $G_{compress}$ and

$G_{predict}$. In this section, we briefly describe how we specify these distributions. We then explain how we compute the optimal clusterings for each language.

2.1. *IB Framework*

The IB framework – the baseline model in the main text – requires specifying two main distributions. The first, $p(m)$, is the need distribution, which captures how frequently each meaning is referred to. As in Carr, Smith, Culbertson, and Kirby (2020) and Imel, Culbertson, Kirby, and Zaslavsky (2025), we assume a uniform need distribution (all 64 stimuli are equally likely to be referred to). The second, $p(u|m)$, defines the mapping from meanings M to referents U, and captures aspects of the semantic space. We define this identically to Imel et al. (2025) (see their paper for further detail).

2.2. *Our Extended Framework*

As in the baseline model above (Imel et al., 2025), our extended framework requires us to specify a need distribution $p(m)$; we use a uniform need distribution as in their work. Our model requires two additional distributions that specify the mappings between meanings and individual feature dimensions: $p(a|m)$, and $p(r|m)$. We assume that these conditional distributions capture deterministic mappings from each meaning to the feature value it has on a particular dimension. For example, the top left Shepard circle in Figure 1 of the main body of the paper has a probability of 1 for referring to the angle 2.566 rad, and a probability of 0 of referring to any other angle.

To compute the optimal set of higher-level features Z_A and Z_R for each language, we considered all possible deterministic partitions of each feature dimension, and found the combination under which a language minimized our objective function. For example, the final system in Chain A (see Figure 1 in the main paper) was optimal under a Z_A which mapped the first two angles to one cluster, the next three angles to a second cluster, and the final three angles to a third cluster (corresponding to the respective rows of the angle-by-radius matrix of feature values in this domain). Its optimal Z_R was the most compressed possible clustering, where all radius values mapped to a single cluster (as the language makes no semantic distinctions along this axis [see the columns of the feature matrix]).

More generally, since there are 8 values per dimension, this leads to roughly 17 million possible partitions along both feature dimensions simultaneously. To make this analysis more computationally feasible, we focused on the exhaustive set of $21,718$ partition combinations that could be described using four words (i.e., languages that make no more than 4 semantic distinctions across the two feature dimensions), as Carr et al. (2020) limited participants to use no more than 4 words in their experimental set-up.

References

Carr, J. W., Smith, K., Culbertson, J., & Kirby, S. (2020). Simplicity and informativeness in semantic category systems. *Cognition*, *202*, 104289.

Imel, N., Culbertson, J., Kirby, S., & Zaslavsky, N. (2025). Iterated language learning is shaped by a drive for optimizing lossy compression. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 47).

Slonim, N., Friedman, N., & Tishby, N. (2006). Multivariate information bottleneck. *Neural computation*, *18*(8), 1739–1789.